

Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence where computers learn patterns from data and make decisions or predictions without being explicitly programmed for every task.

Instead of writing fixed rules, we give the computer:

- **Data**
- **Examples**
- **Algorithms**

The machine learns from the data and improves its performance over time.

Simple Example

If we show a computer thousands of emails marked as **spam** or **not spam**, it learns the pattern and can automatically identify spam emails in the future.

Types of Machine Learning

Machine Learning is mainly divided into **4 types**.

1. Supervised Learning

In supervised learning, the machine is trained using **labeled data**.

Labeled data means:

- Input data already has the correct output.

The model learns the relationship between input and output.

Example

Input	Output
-------	--------

Study hours	Exam marks
-------------	------------

Email text	Spam / Not Spam
------------	-----------------

Applications

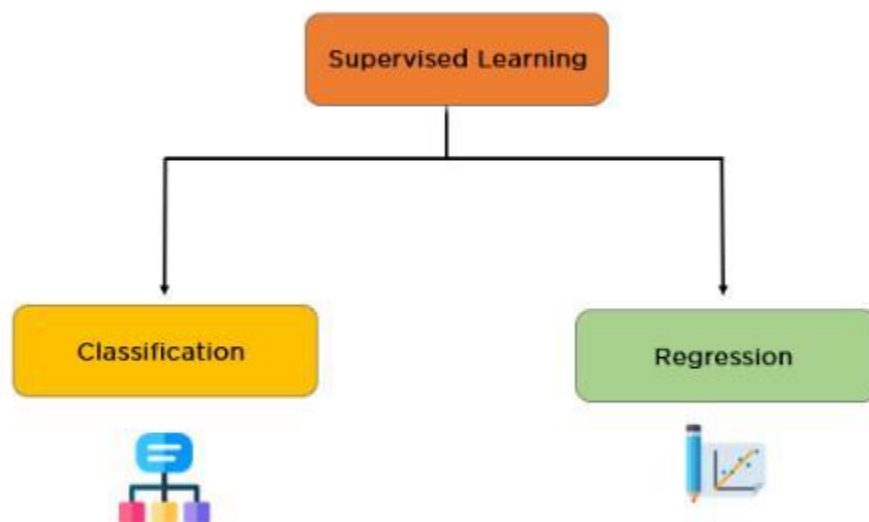
- Email spam detection
- Weather prediction

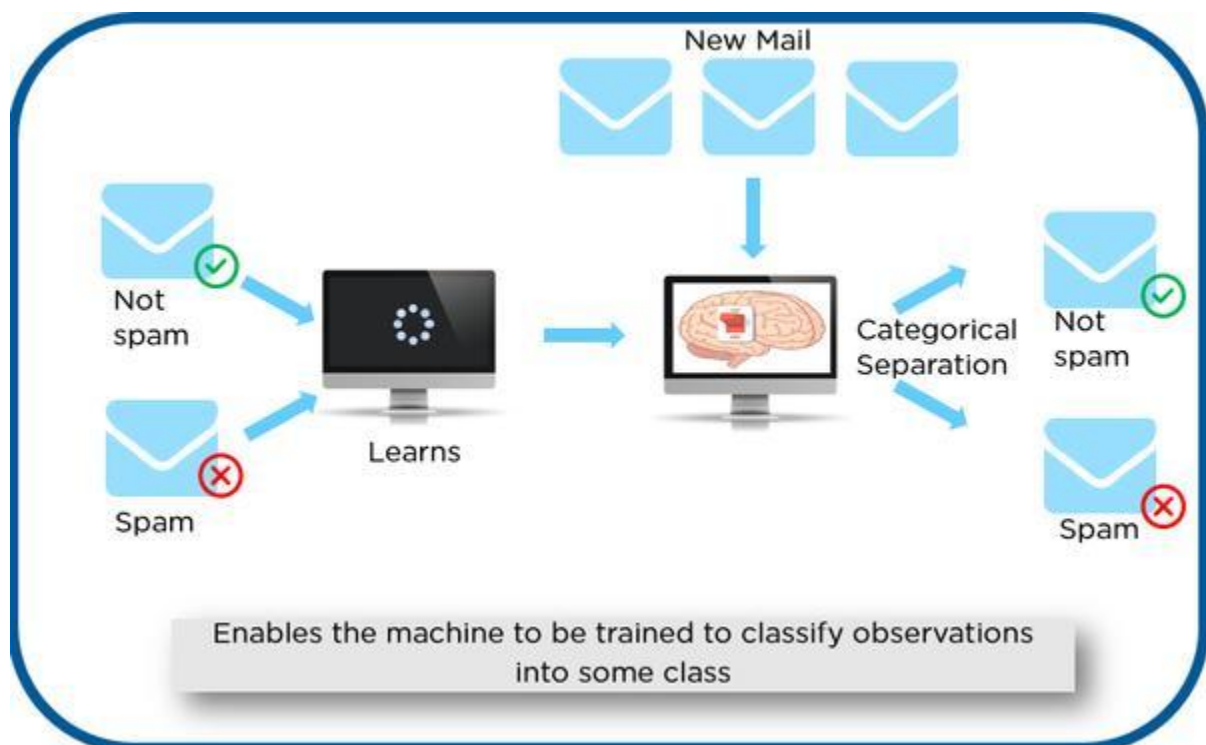
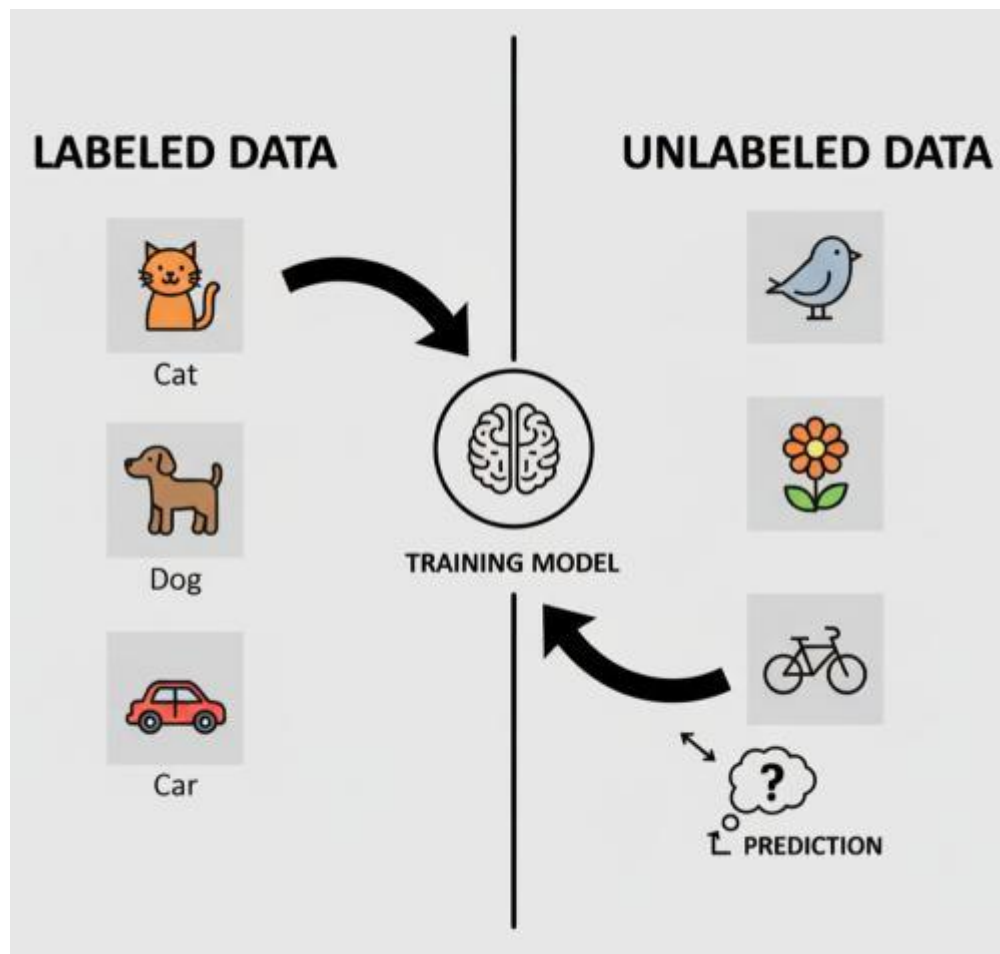
- House price prediction
- Image classification

Common Algorithms

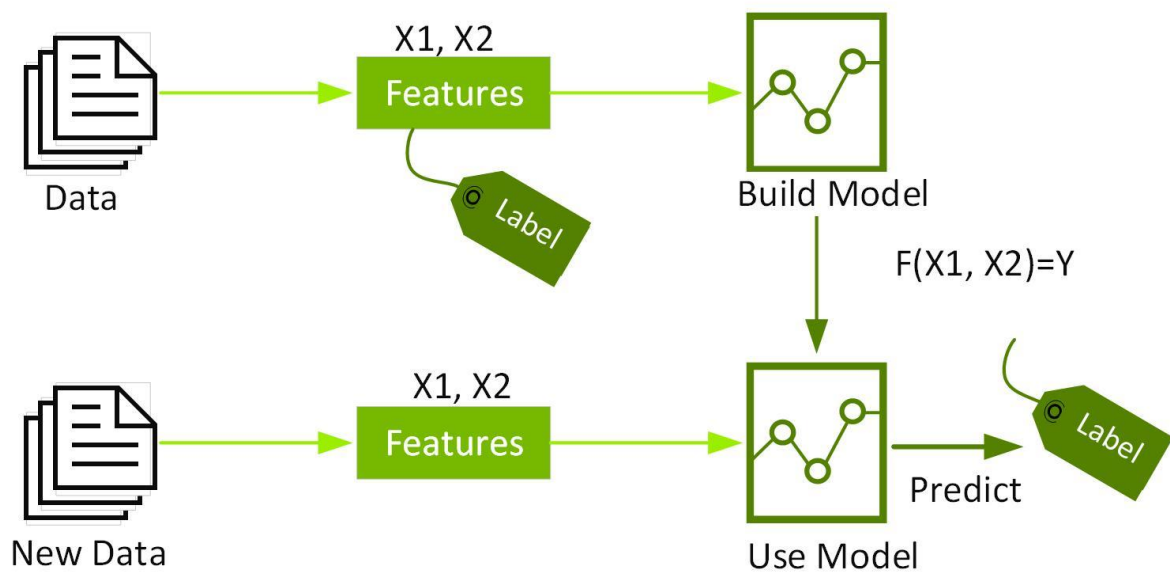
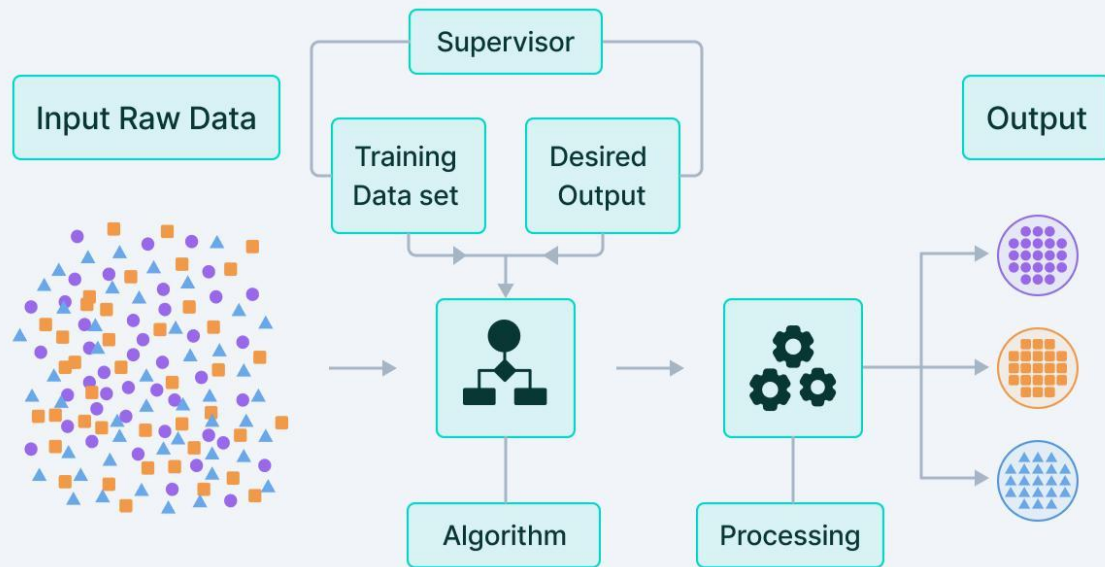
- Linear Regression
- Logistic Regression
- Decision Tree
- Support Vector Machine (SVM)

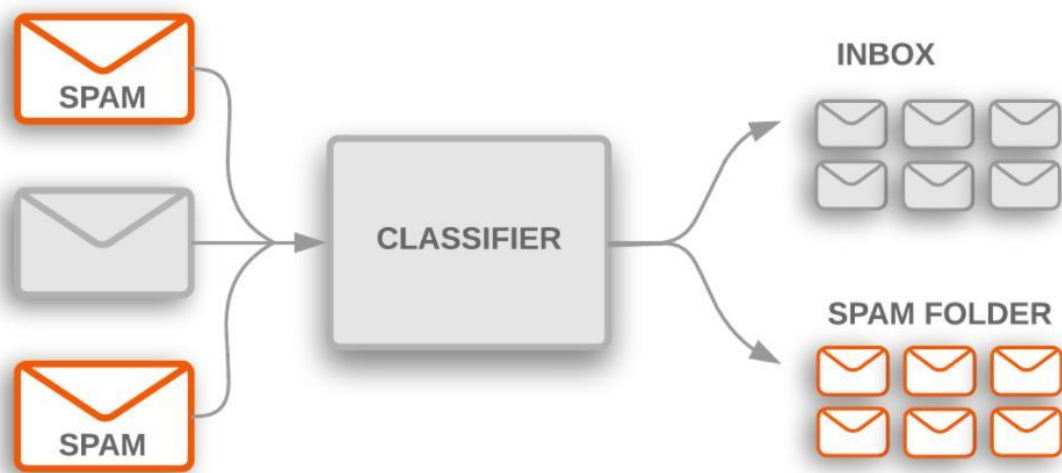
Diagram





Supervised Learning





2. Unsupervised Learning

In unsupervised learning, the machine is trained using **unlabeled data**.

The system tries to find hidden patterns or groups by itself.

Example

A shopping app groups customers based on buying behavior.

Applications

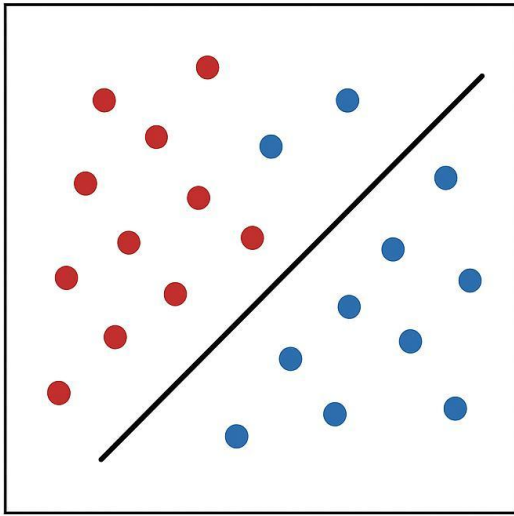
- Customer segmentation
- Recommendation systems
- Pattern detection
- Data grouping

Common Algorithms

- K-Means Clustering
- Hierarchical Clustering
- PCA (Principal Component Analysis)

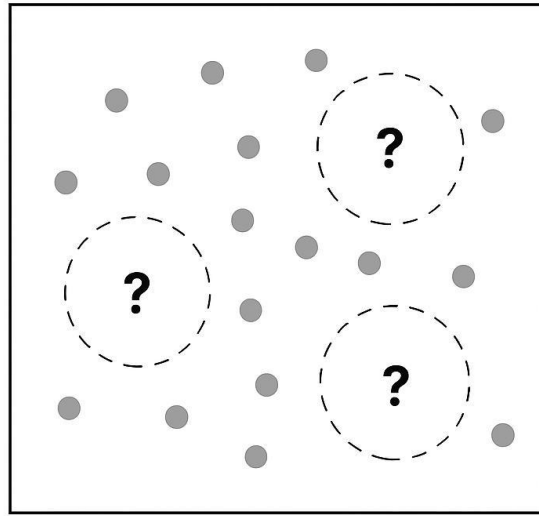
Diagram

Supervised



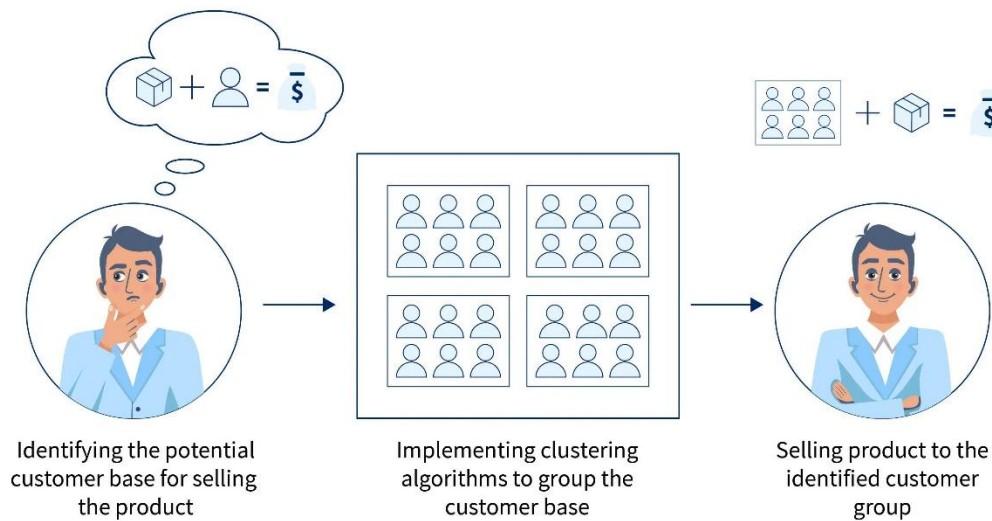
Known Classes

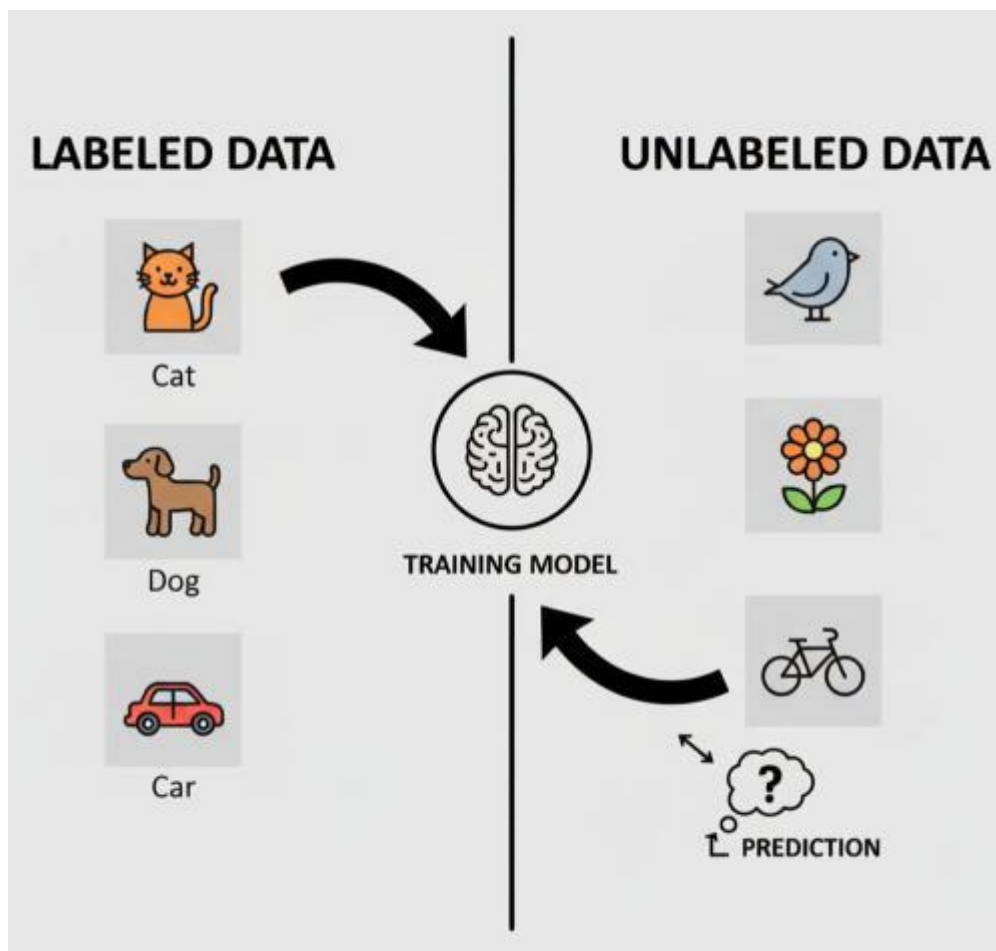
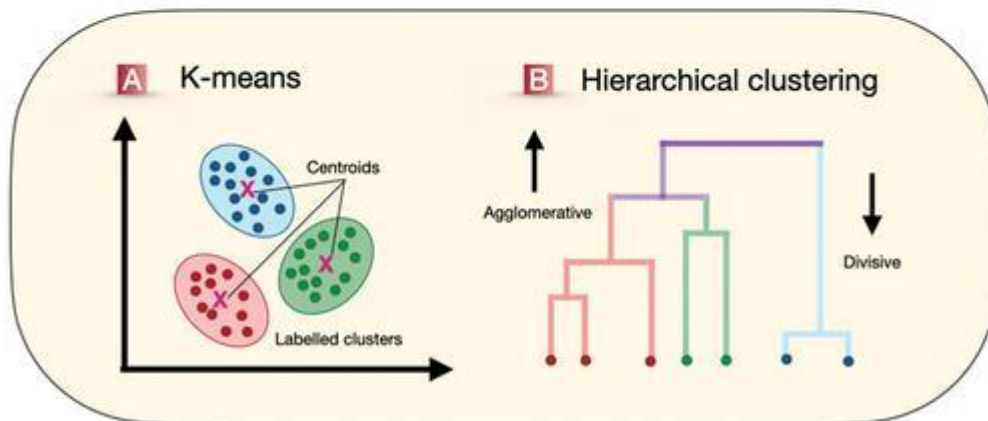
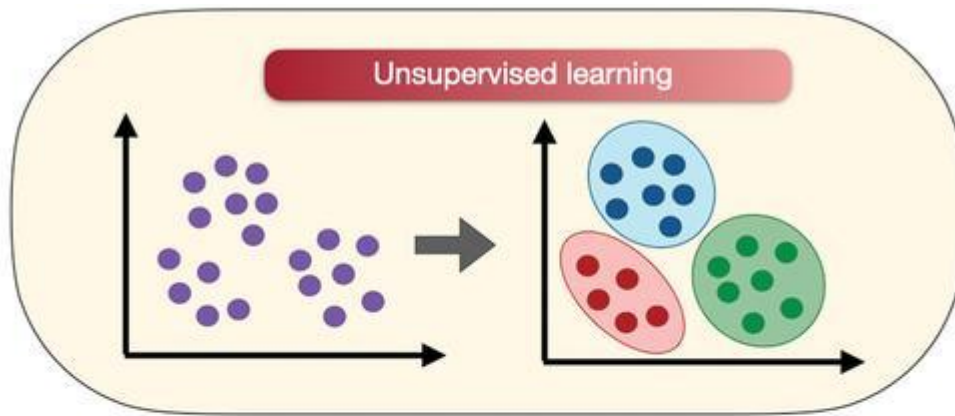
Unsupervised

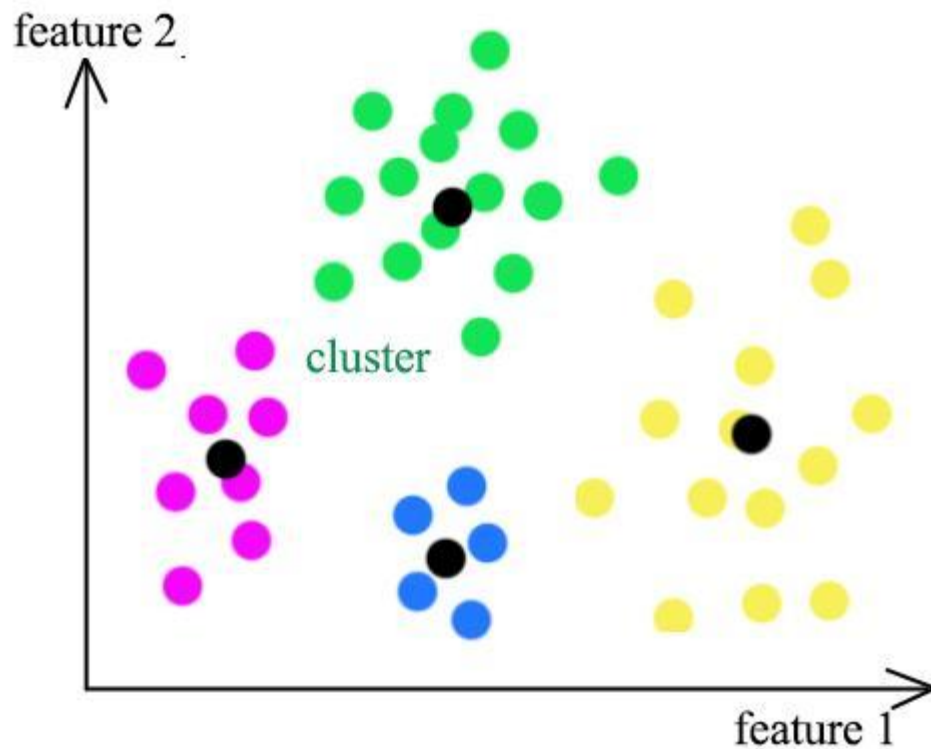


Hidden Patterns to Discover

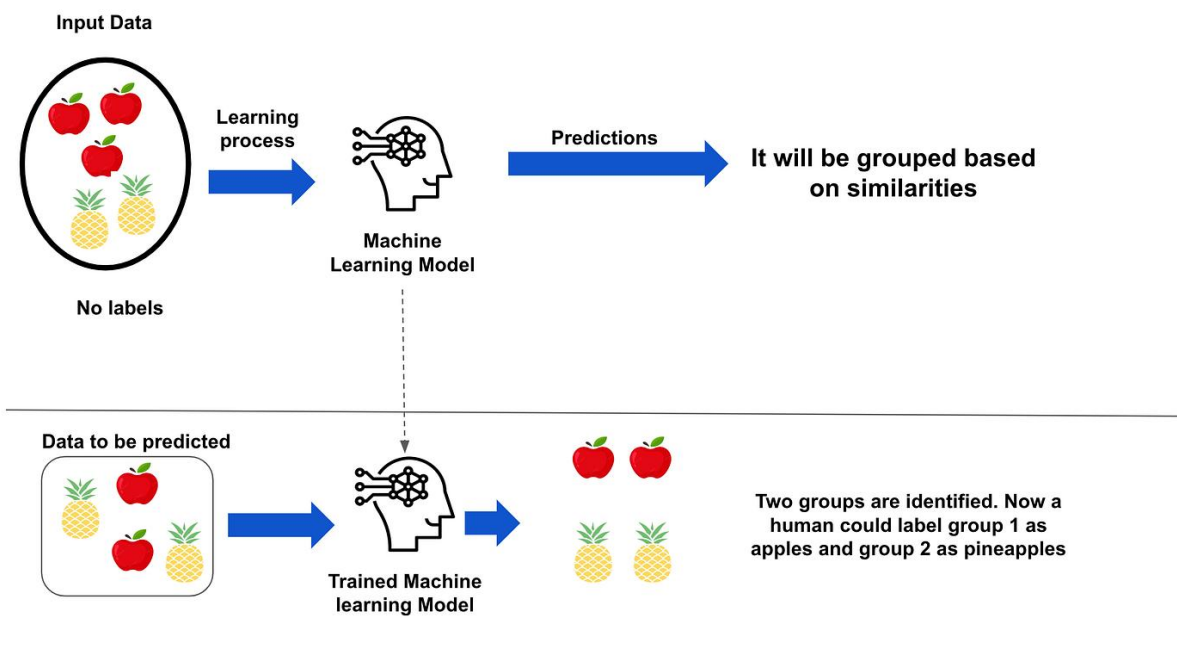
Machine Learning Project - Customer Segmentation







Unsupervised Learning



3. Reinforcement Learning

In reinforcement learning, an agent learns by:

- Performing actions

- Receiving rewards or penalties

The goal is to maximize the reward.

Example

A robot learns to walk by trying different movements.

Applications

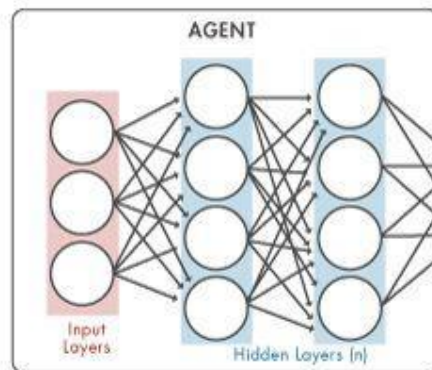
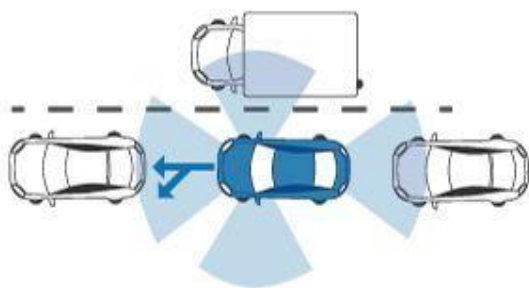
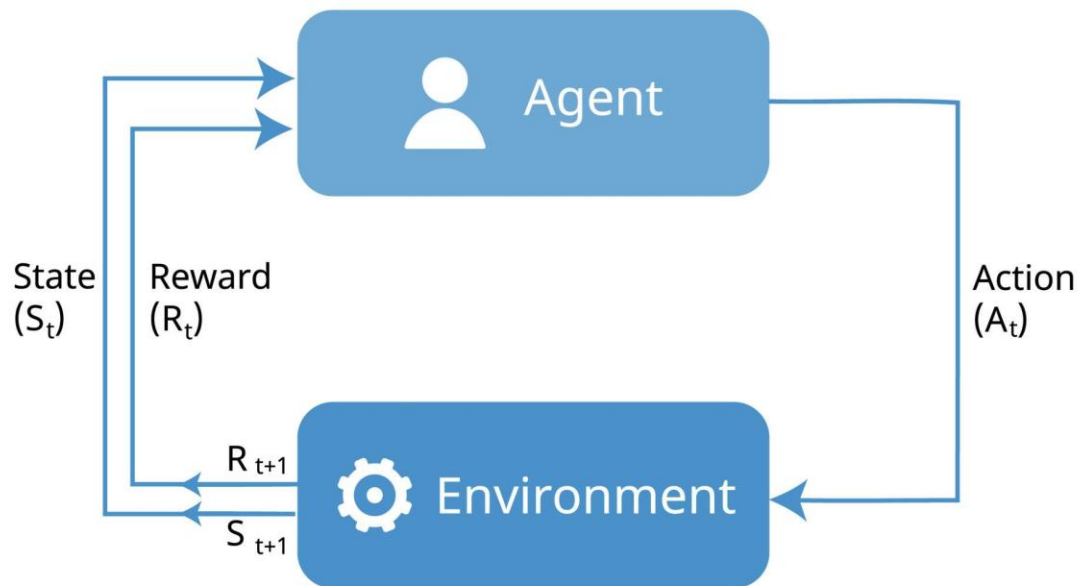
- Self-driving cars
- Robotics
- Game playing
- AI assistants

Key Terms

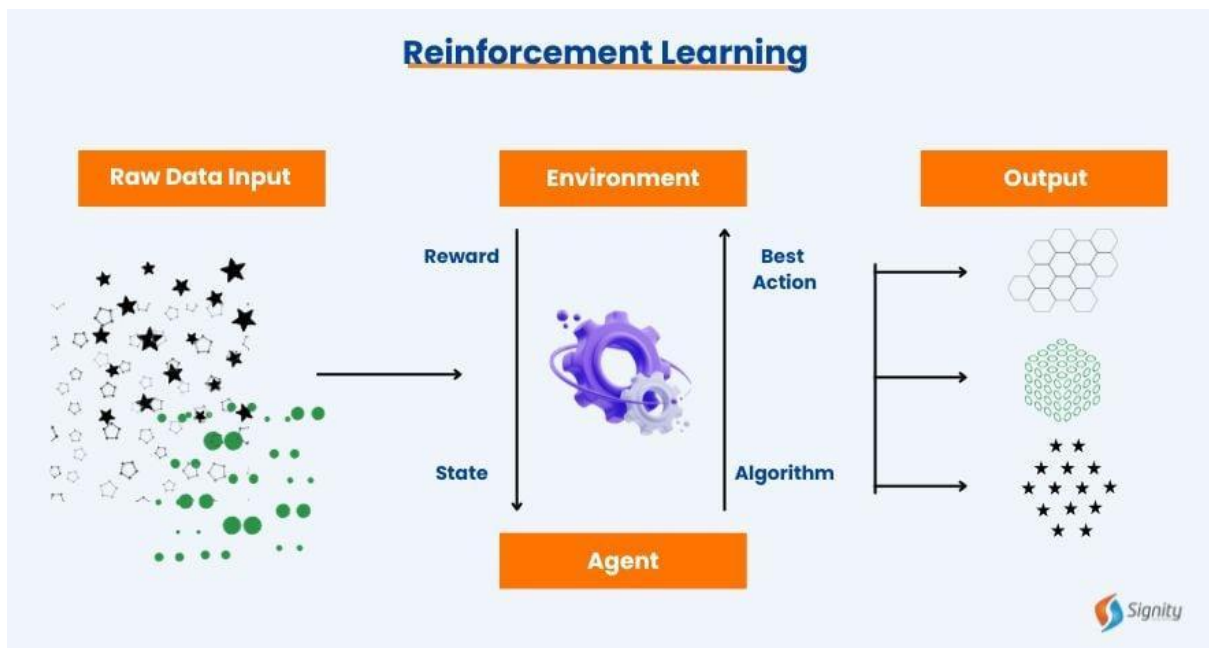
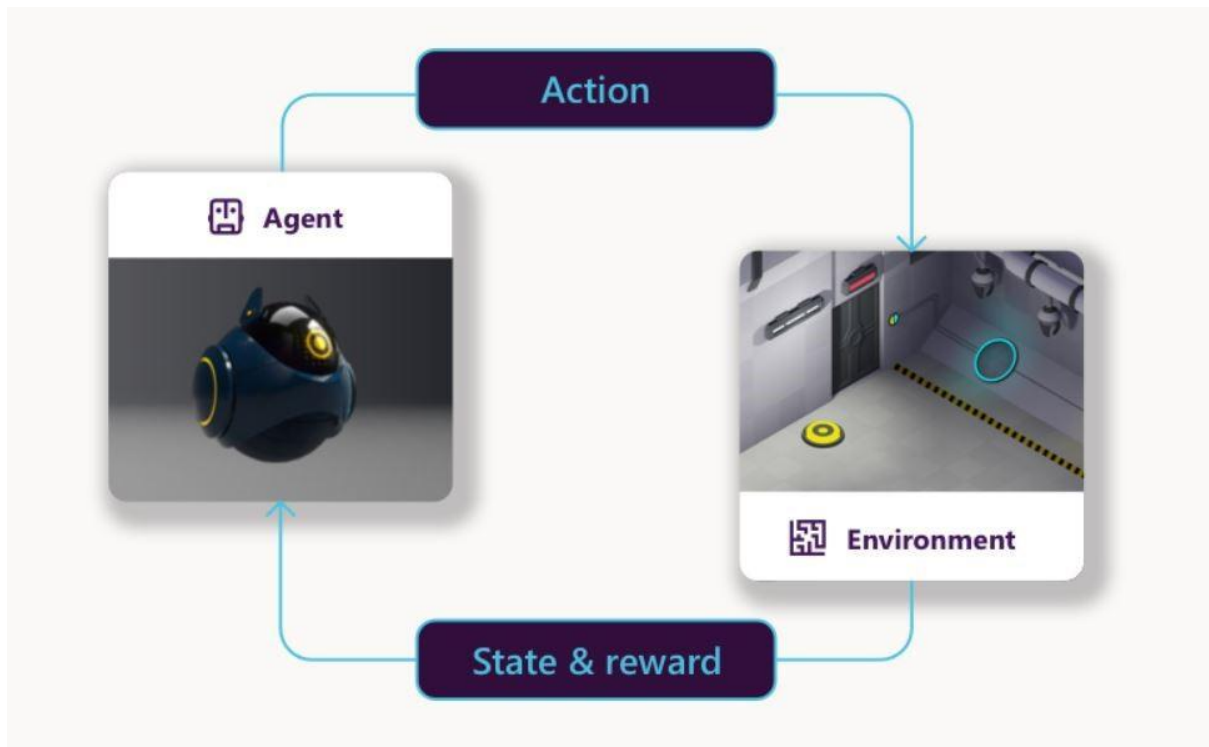
- Agent
- Environment
- Reward
- Action

Diagram

Reinforcement Learning



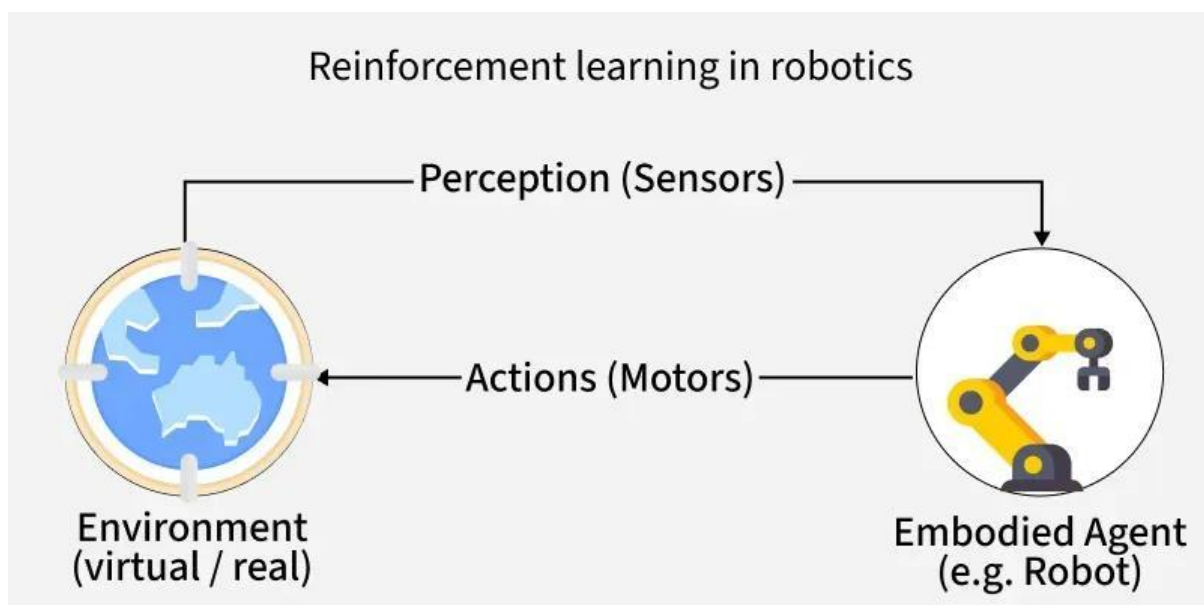
- TURN LEFT
- TURN RIGHT
- BRAKE
- ACCELERATE



Reinforcement Learning Workflow



18



4. Semi-Supervised Learning

This combines:

- A small amount of labeled data
- A large amount of unlabeled data

It is useful when labeling data is expensive.

Applications

- Medical image analysis

- Speech recognition
- Web content classification

Diagram

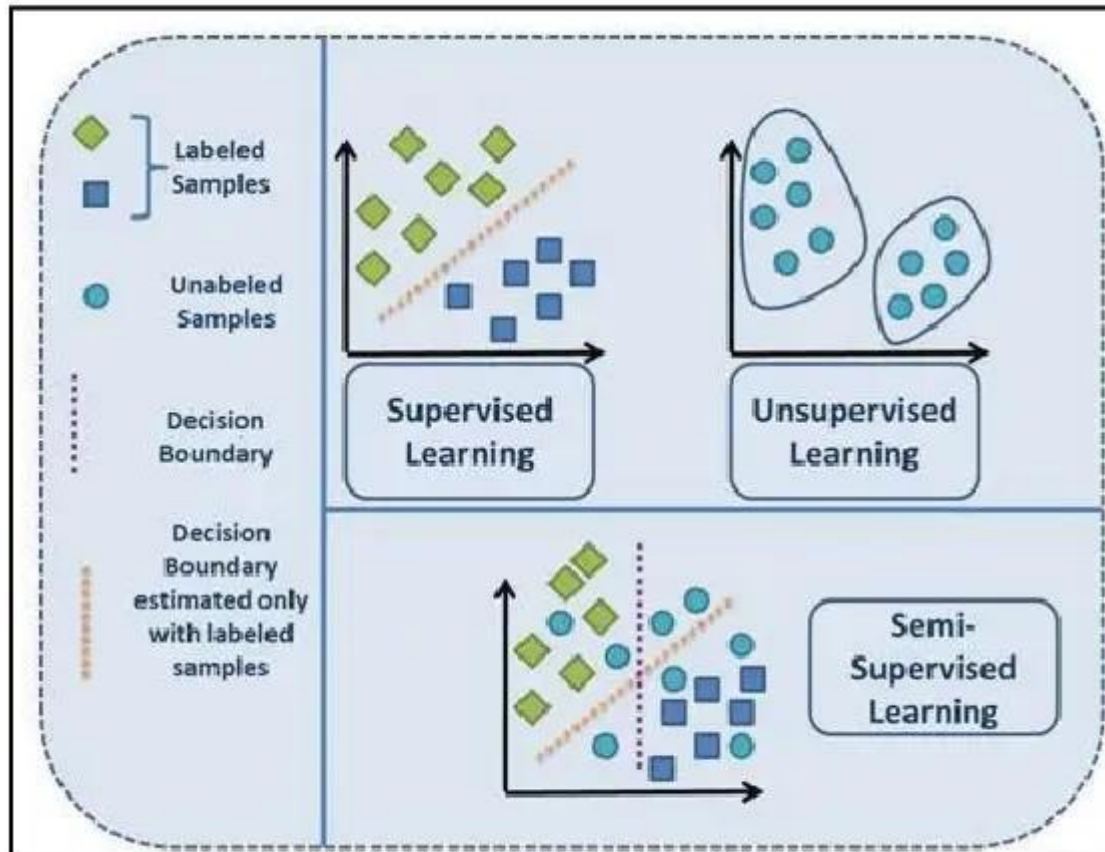
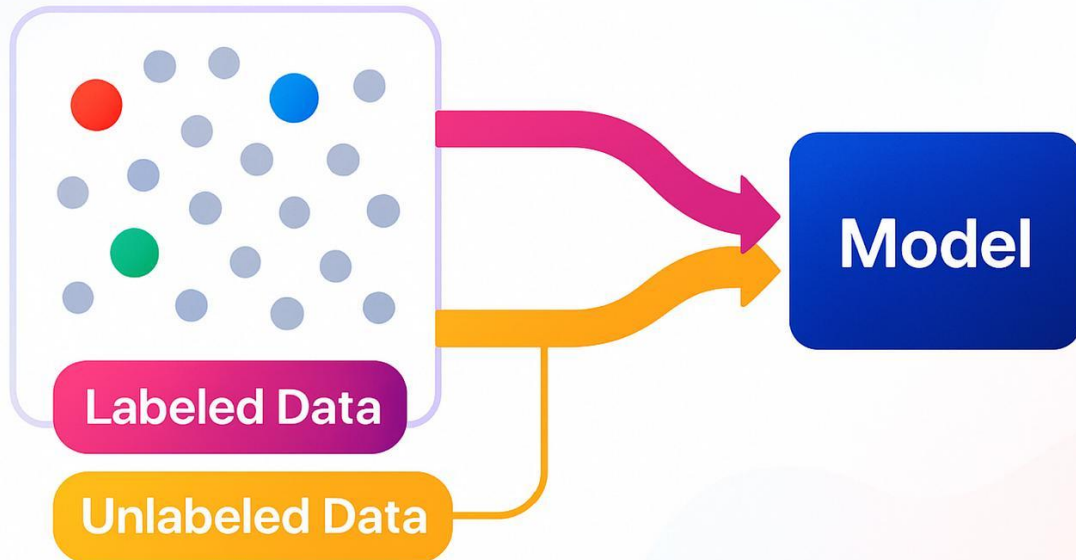


Fig. 1 Semi-Supervised Learning

Semi-supervised Learning



G2.com

Labeled data

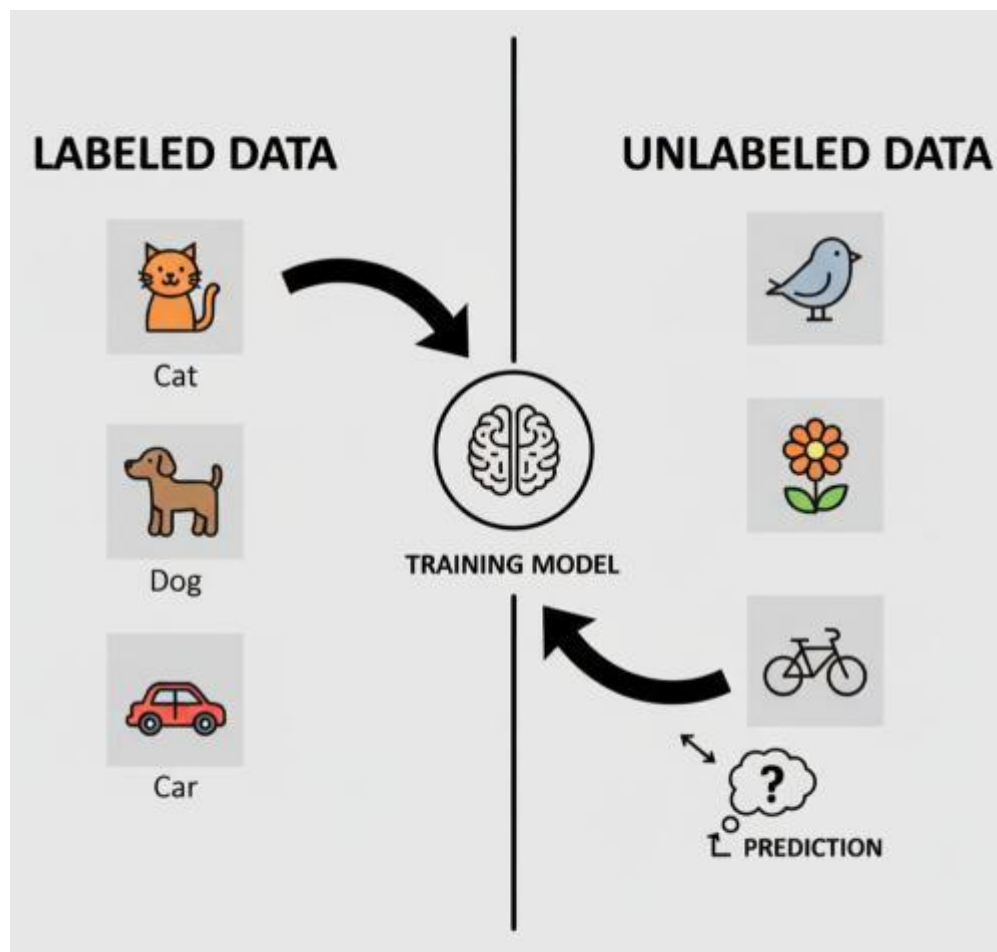
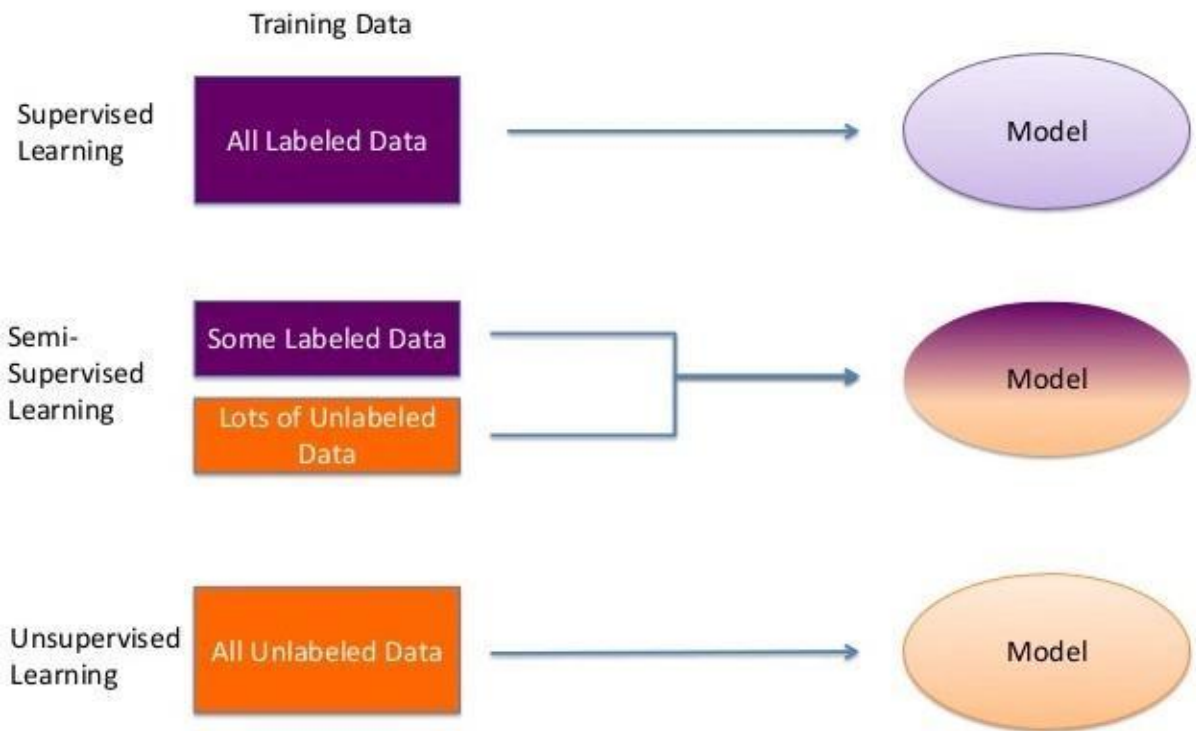


Tagged and used in supervised learning.

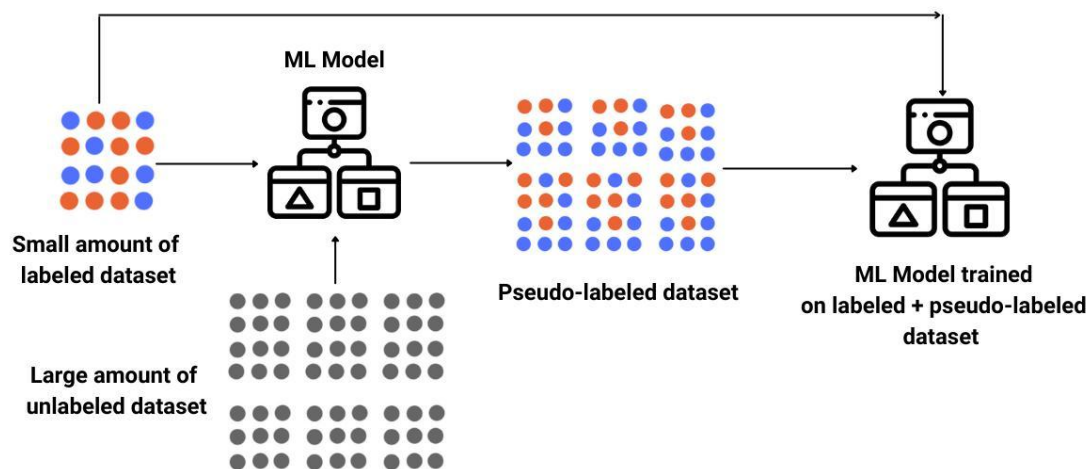
Unlabeled data



Untagged and used in unsupervised learning.



Semi-supervised learning use-case



Comparison of Types

Type	Data Used	Main Goal
Supervised Learning	Labeled Data	Predict output
Unsupervised Learning	Unlabeled Data	Find patterns
Reinforcement Learning	Rewards & Penalties	Learn best action
Semi-Supervised Learning	Mixed Data	Improve accuracy with less labeled data

Advantages of Machine Learning

- Automates tasks
- Improves accuracy
- Handles large data
- Learns and improves over time

Disadvantages

- Requires large data
- Can be expensive
- Sometimes difficult to understand predictions

- Needs powerful computers for complex tasks
-

Real-Life Examples

- Netflix recommendations
 - Google Translate
 - Face recognition
 - Chatbots
 - Online shopping suggestions
-

Conclusion

Machine Learning enables computers to learn from data and make intelligent decisions. Different types of machine learning are used depending on the nature of the problem and the available data.

Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence where computers learn patterns from data and make decisions or predictions without being explicitly programmed for every task.

Instead of writing fixed rules, we give the computer:

- **Data**
- **Examples**
- **Algorithms**

The machine learns from the data and improves its performance over time.

Simple Example

If we show a computer thousands of emails marked as **spam** or **not spam**, it learns the pattern and can automatically identify spam emails in the future.

Types of Machine Learning

Machine Learning is mainly divided into **4 types**.

1. Supervised Learning

In supervised learning, the machine is trained using **labeled data**.

Labeled data means:

- Input data already has the correct output.

The model learns the relationship between input and output.

Example

Input	Output
-------	--------

Study hours	Exam marks
-------------	------------

Email text	Spam / Not Spam
------------	-----------------

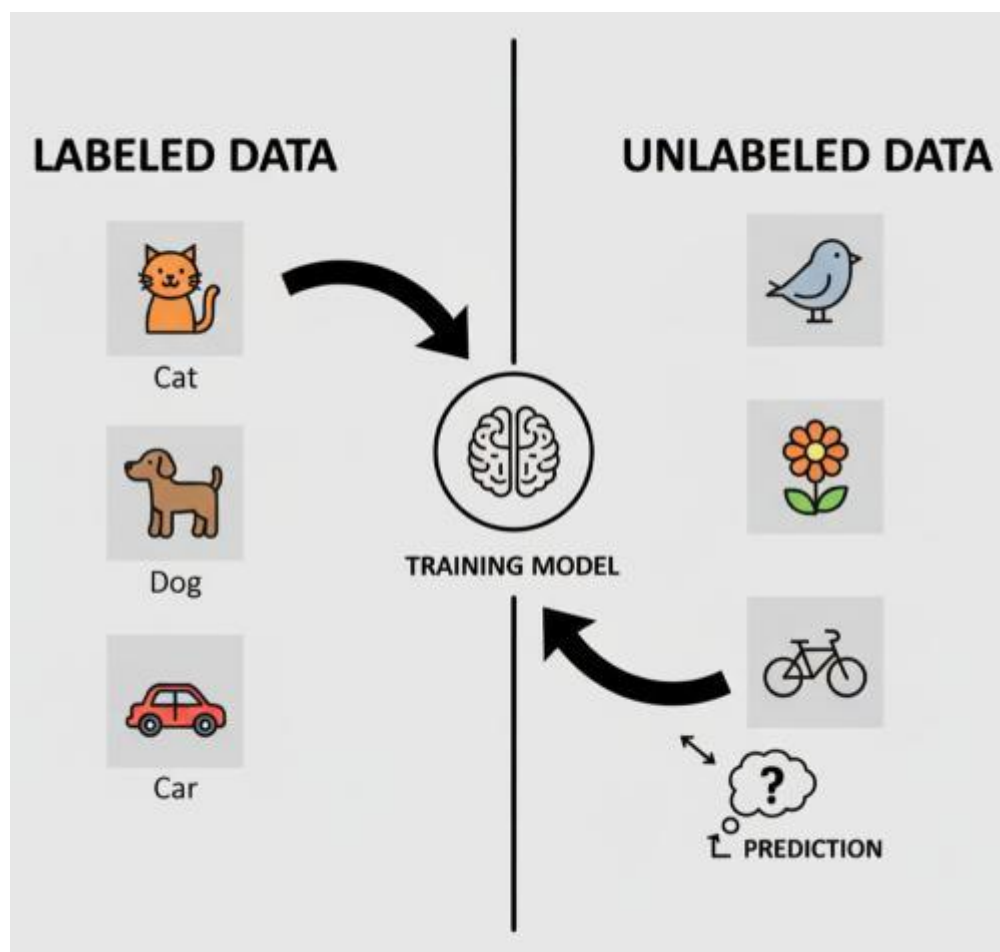
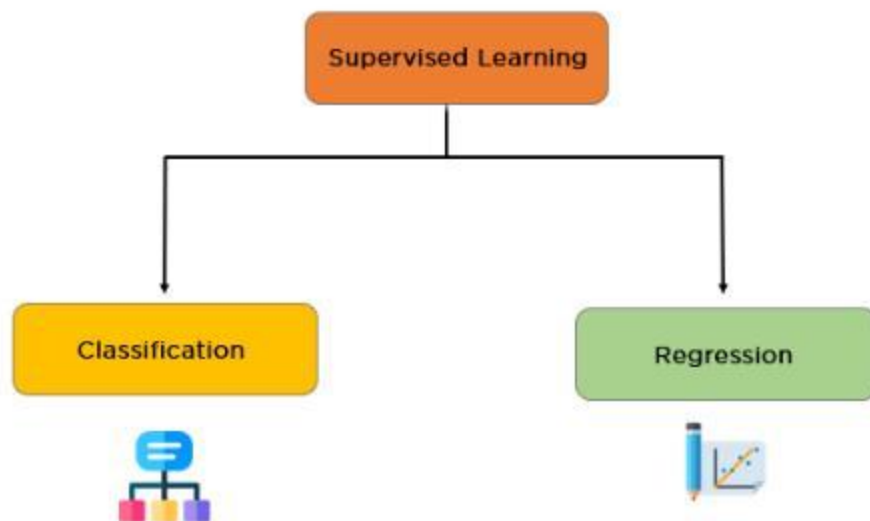
Applications

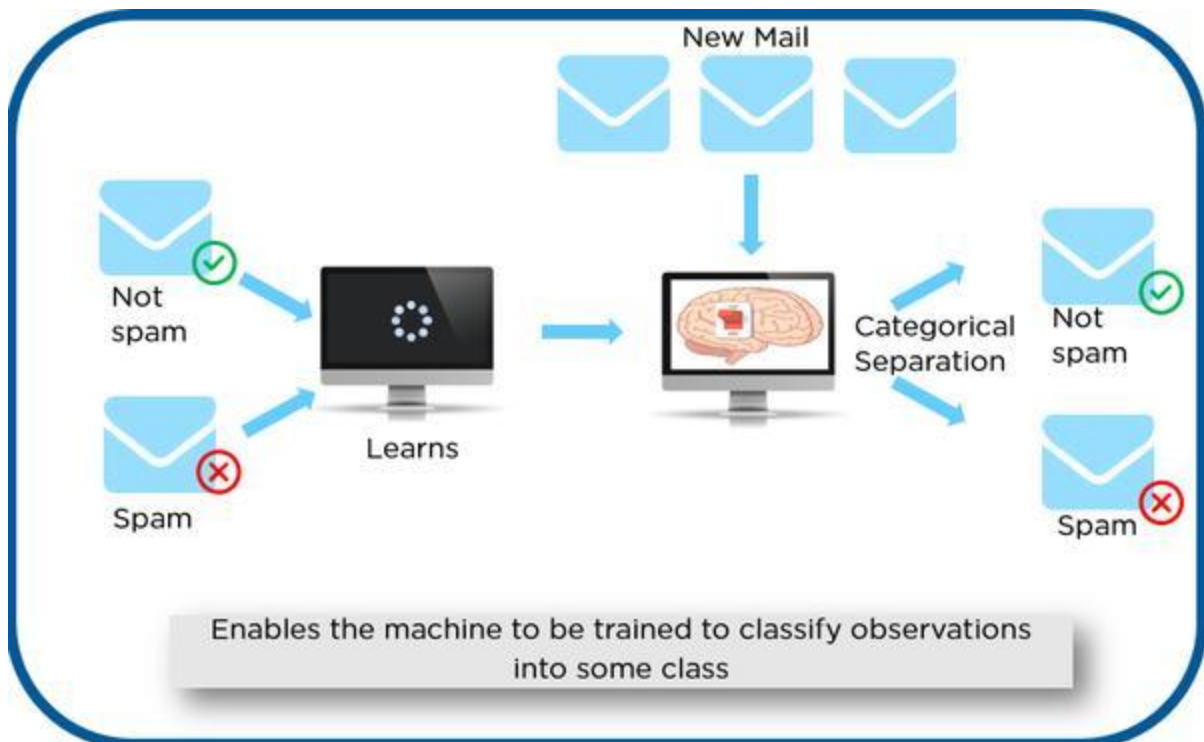
- Email spam detection
- Weather prediction
- House price prediction
- Image classification

Common Algorithms

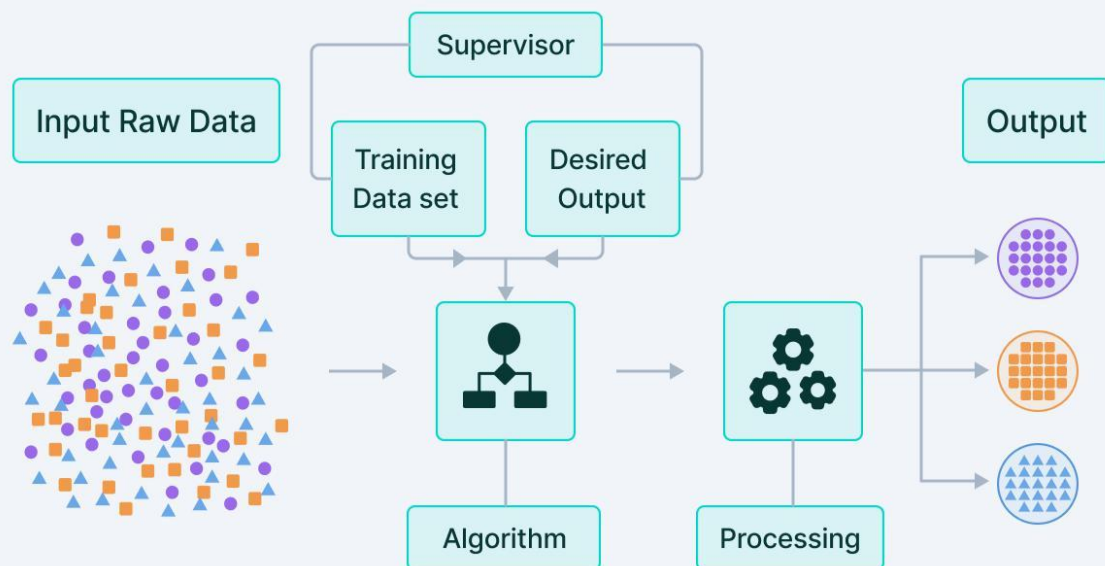
- Linear Regression
- Logistic Regression
- Decision Tree
- Support Vector Machine (SVM)

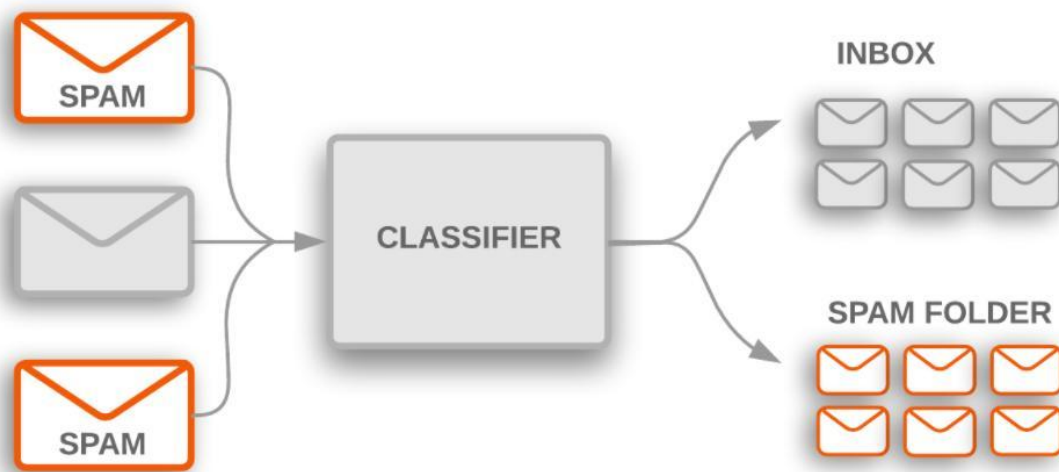
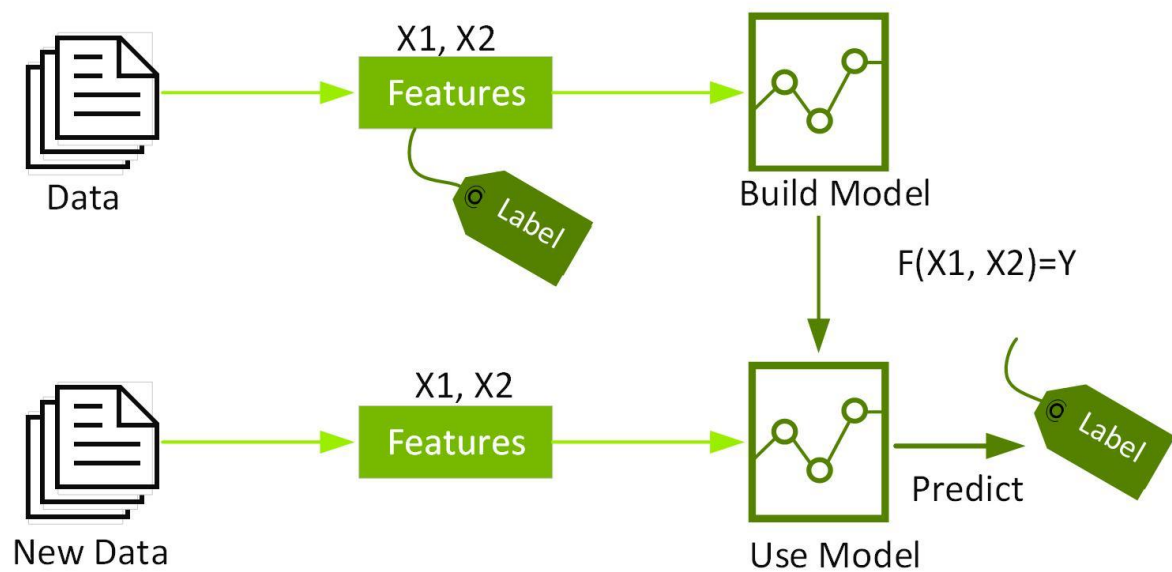
Diagram





Supervised Learning





2. Unsupervised Learning

In unsupervised learning, the machine is trained using **unlabeled data**.

The system tries to find hidden patterns or groups by itself.

Example

A shopping app groups customers based on buying behavior.

Applications

- Customer segmentation

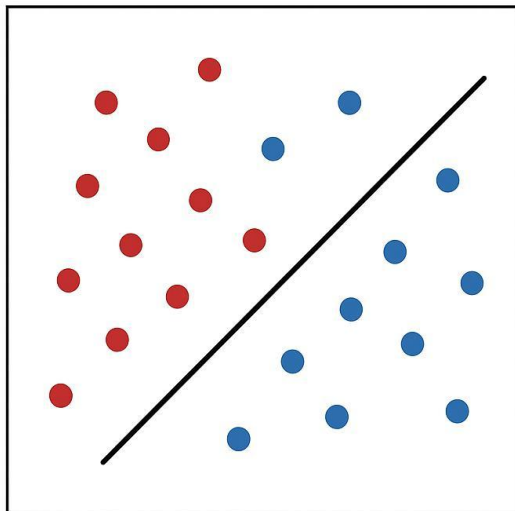
- Recommendation systems
- Pattern detection
- Data grouping

Common Algorithms

- K-Means Clustering
- Hierarchical Clustering
- PCA (Principal Component Analysis)

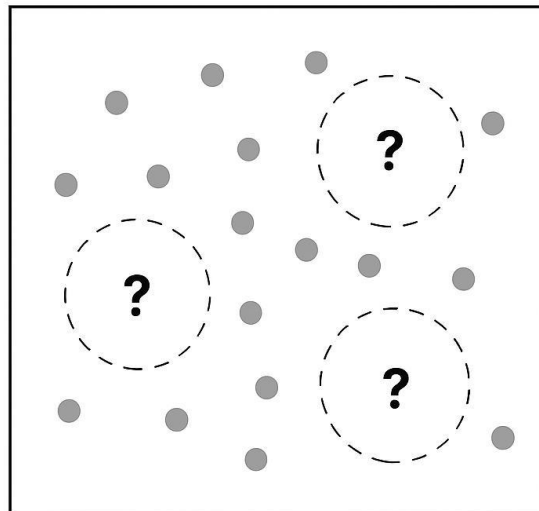
Diagram

Super vised



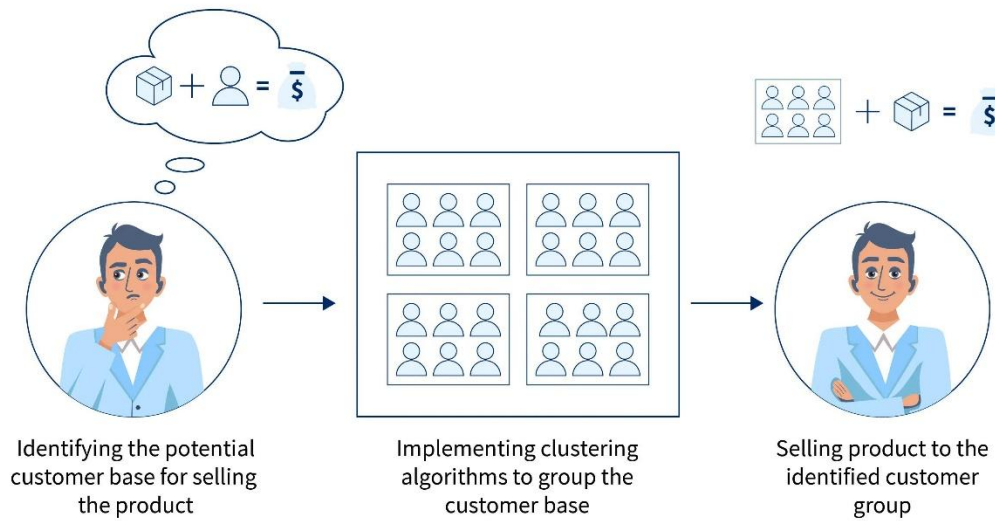
Known Classes

Unsupervised

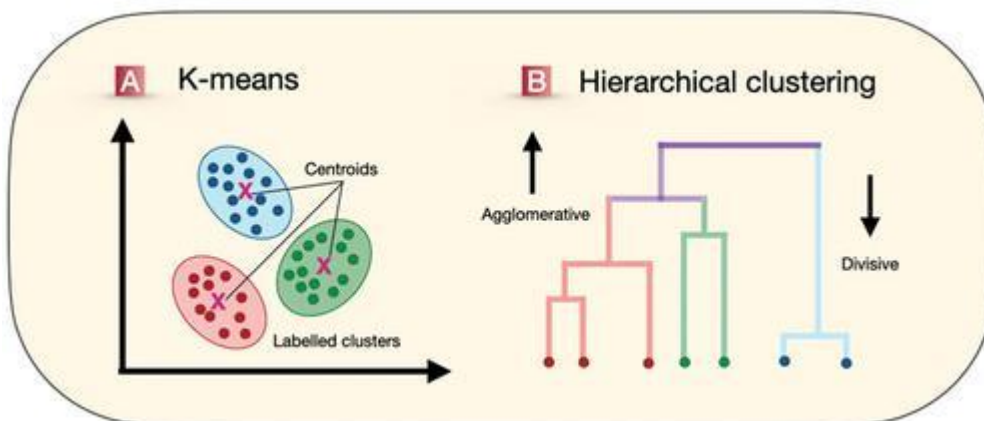
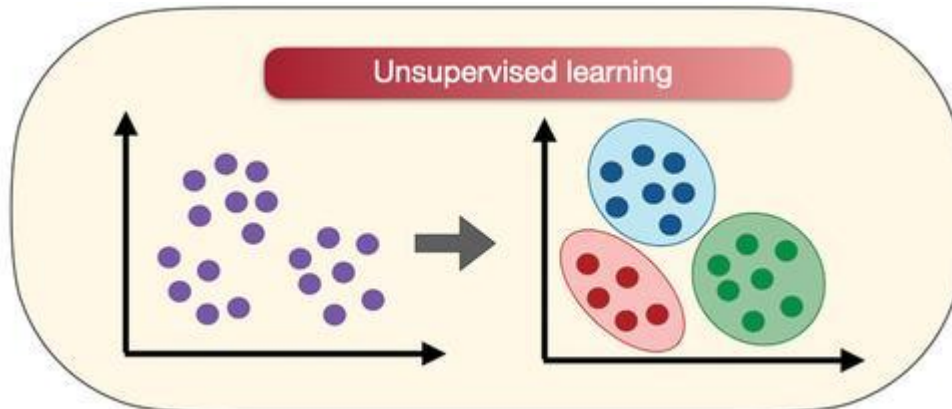


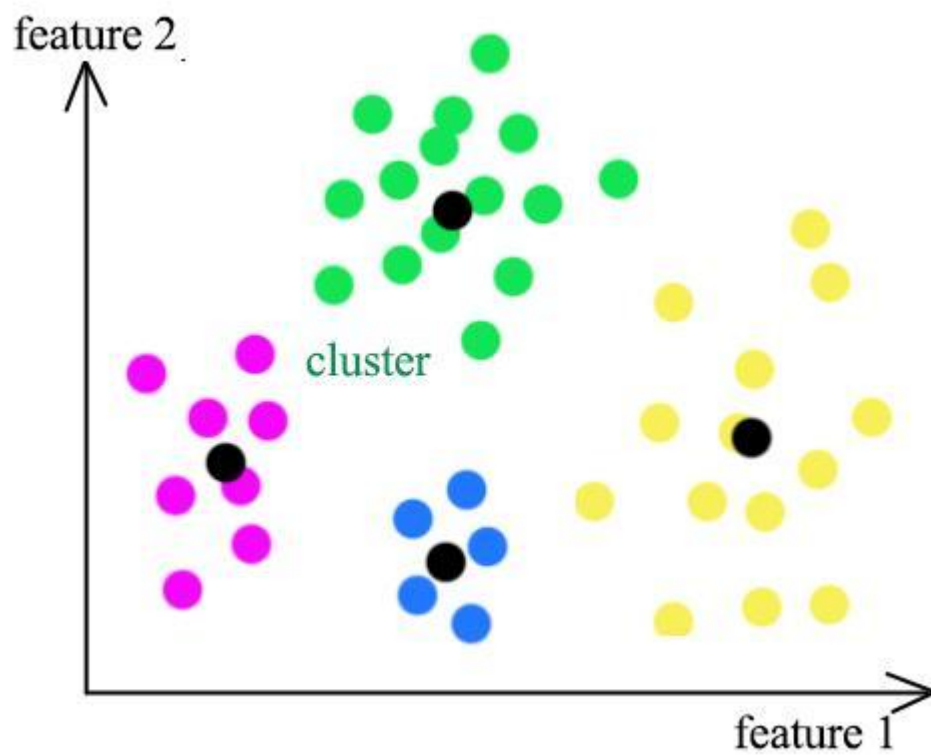
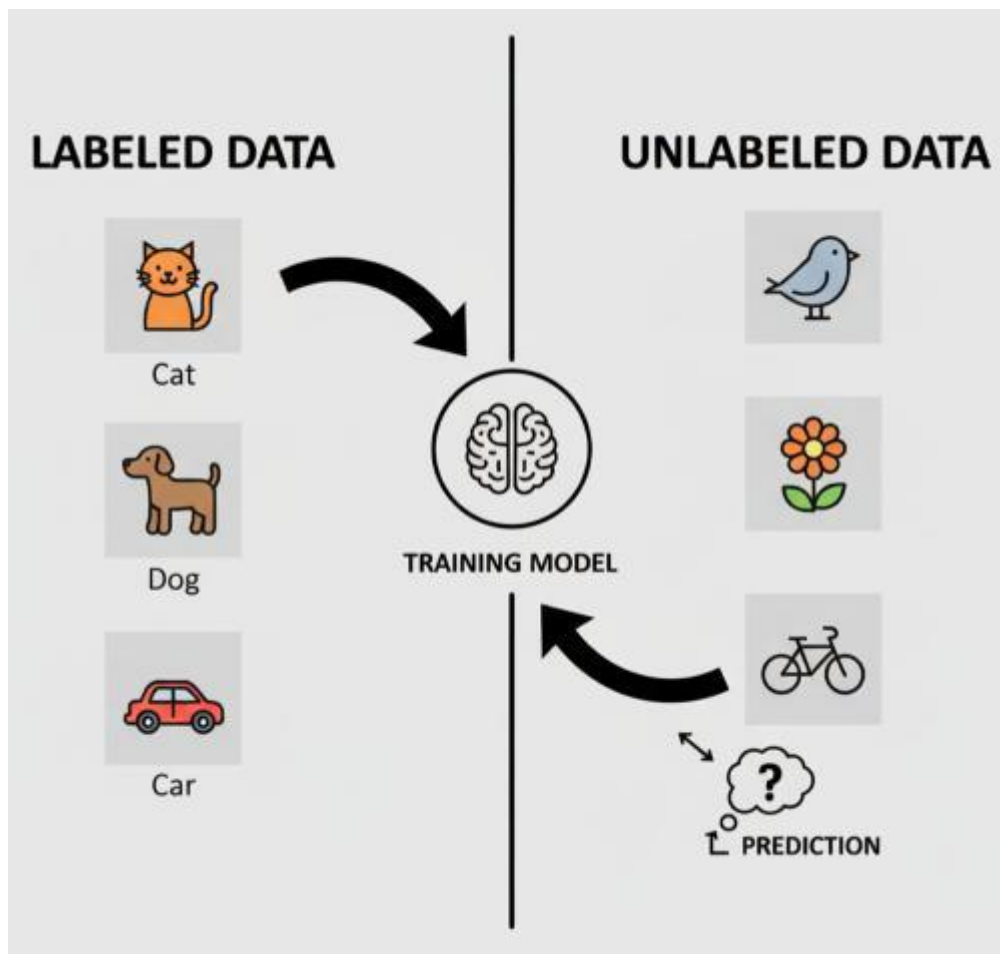
Hidden Patterns to Discover

Machine Learning Project - Customer Segmentation

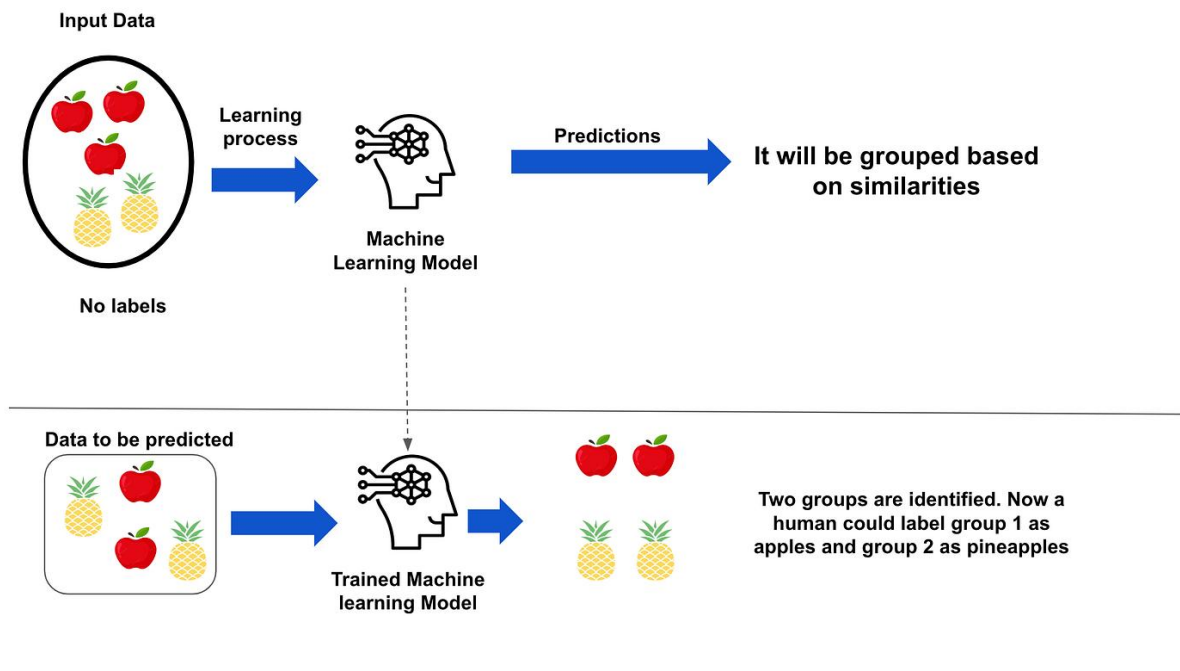


SCALER
Topics





Unsupervised Learning



3. Reinforcement Learning

In reinforcement learning, an agent learns by:

- Performing actions
- Receiving rewards or penalties

The goal is to maximize the reward.

Example

A robot learns to walk by trying different movements.

Applications

- Self-driving cars
- Robotics
- Game playing
- AI assistants

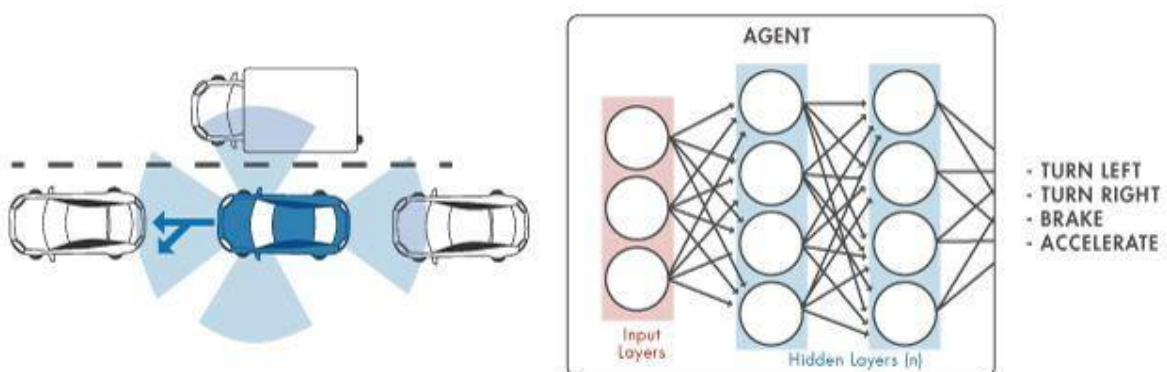
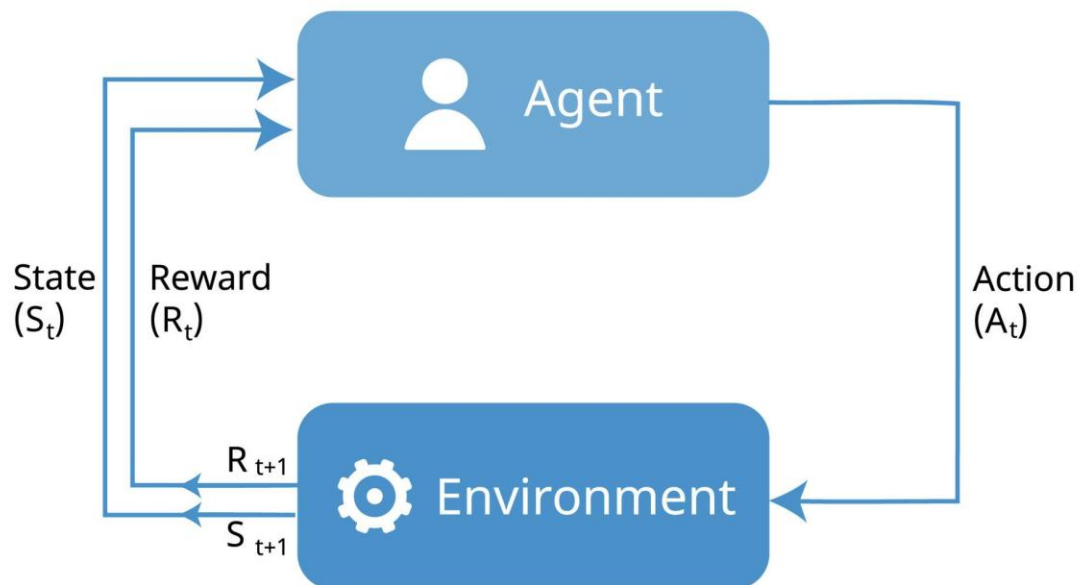
Key Terms

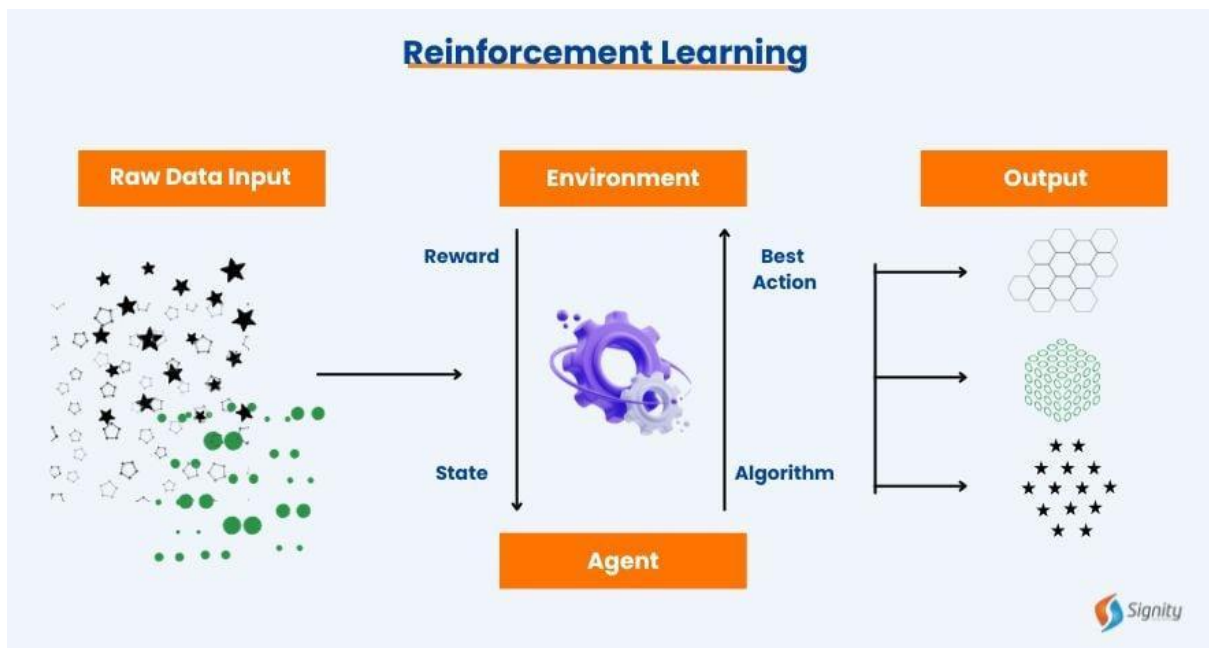
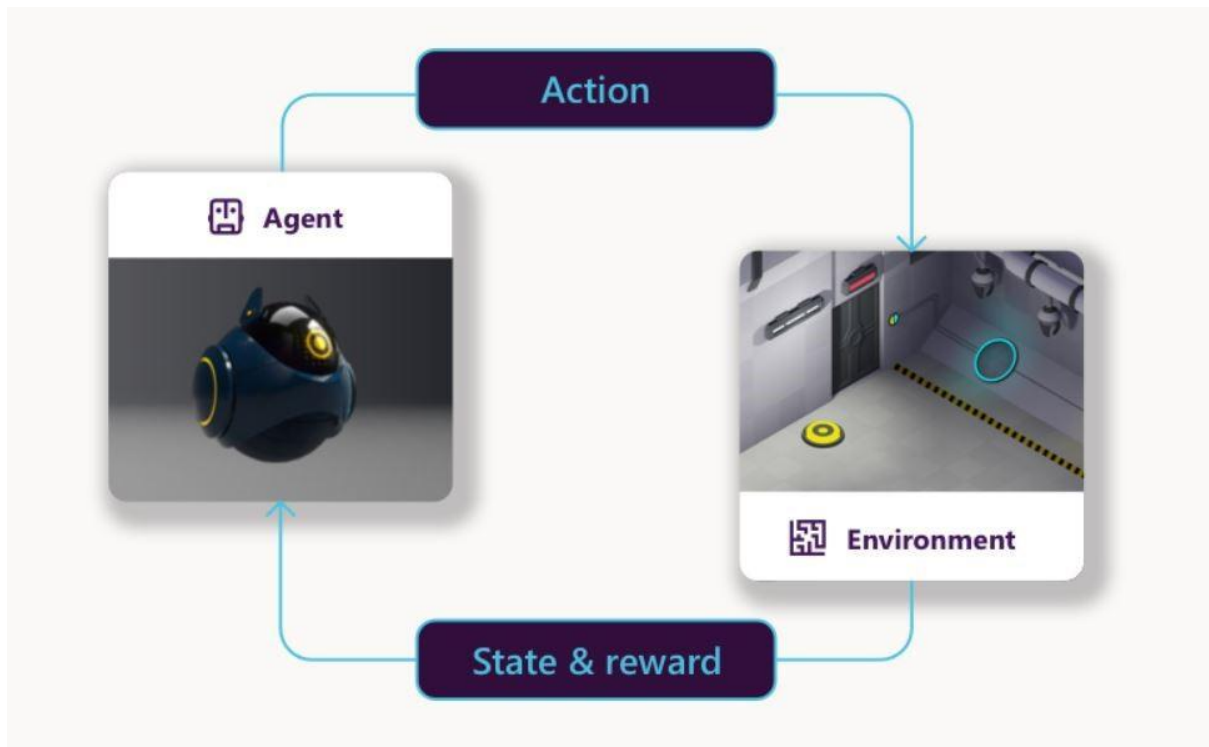
- Agent
- Environment
- Reward

- Action

Diagram

Reinforcement Learning

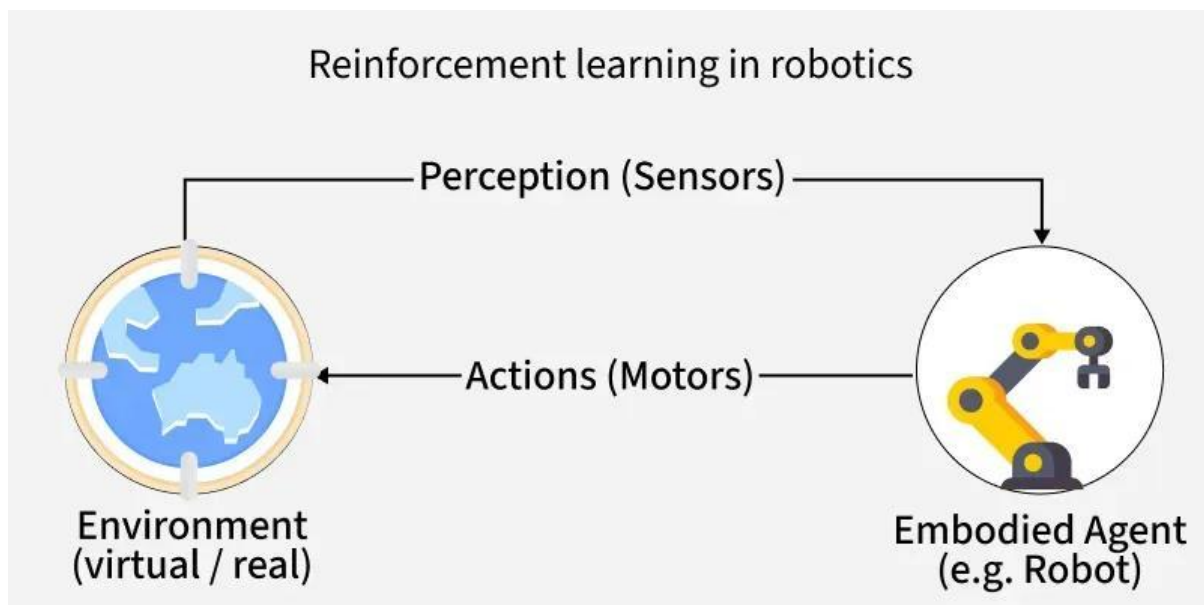




Reinforcement Learning Workflow



18



4. Semi-Supervised Learning

This combines:

- A small amount of labeled data
- A large amount of unlabeled data

It is useful when labeling data is expensive.

Applications

- Medical image analysis

- Speech recognition
- Web content classification

Diagram

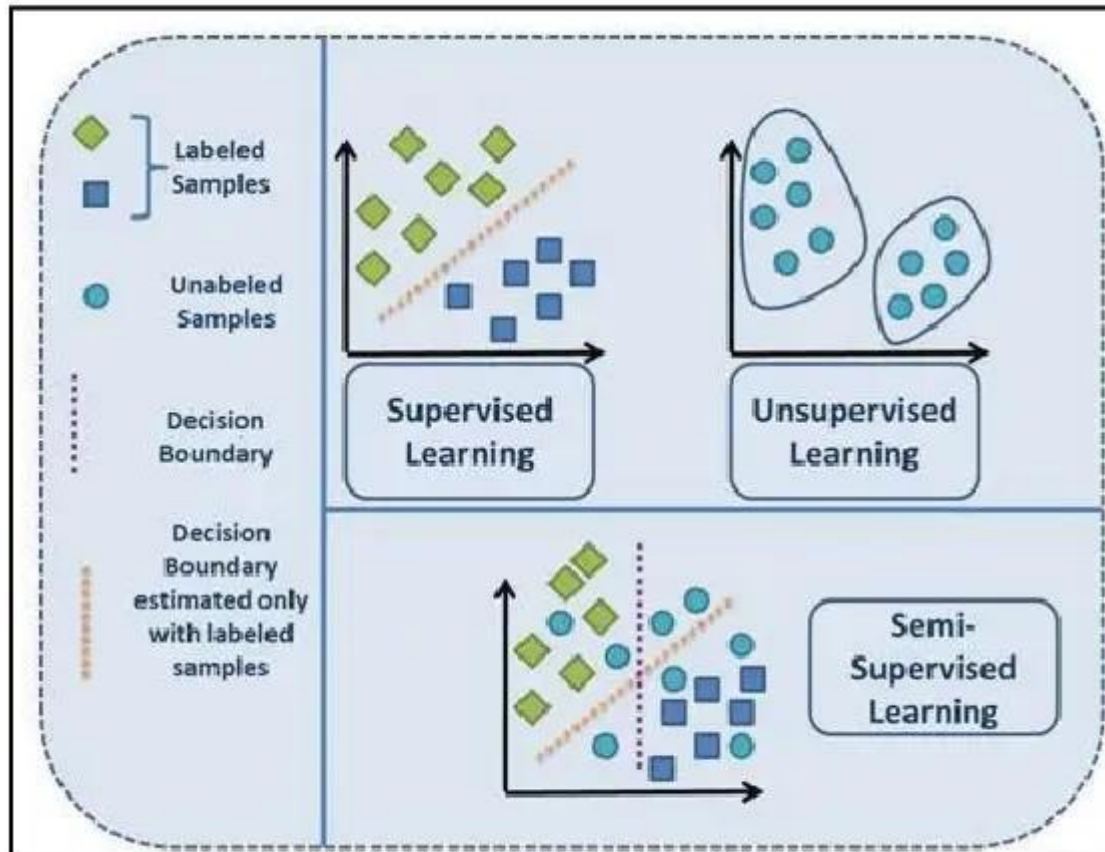
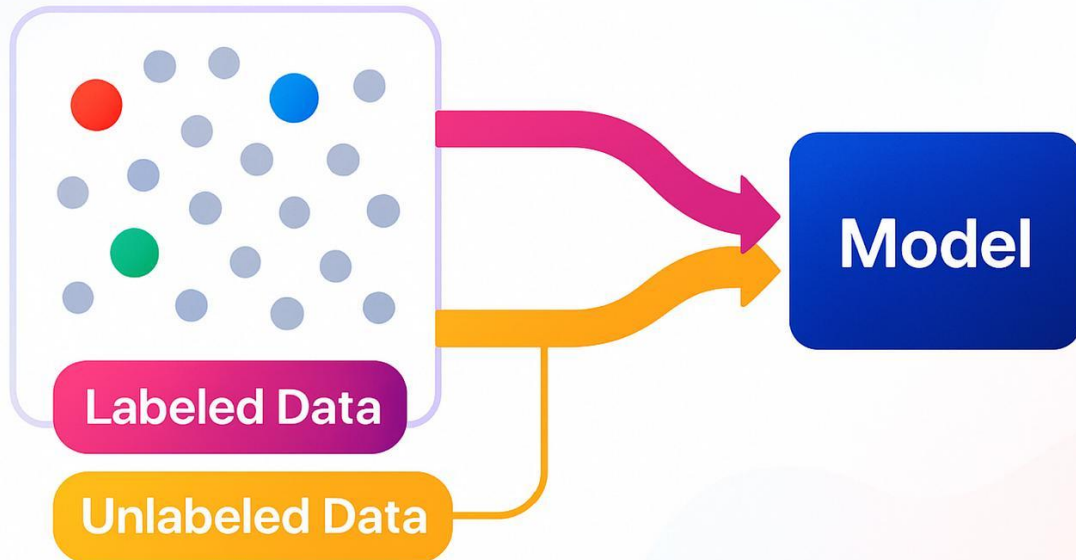


Fig. 1 Semi-Supervised Learning

Semi-supervised Learning



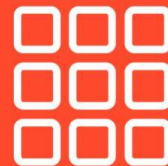
G2.com

Labeled data

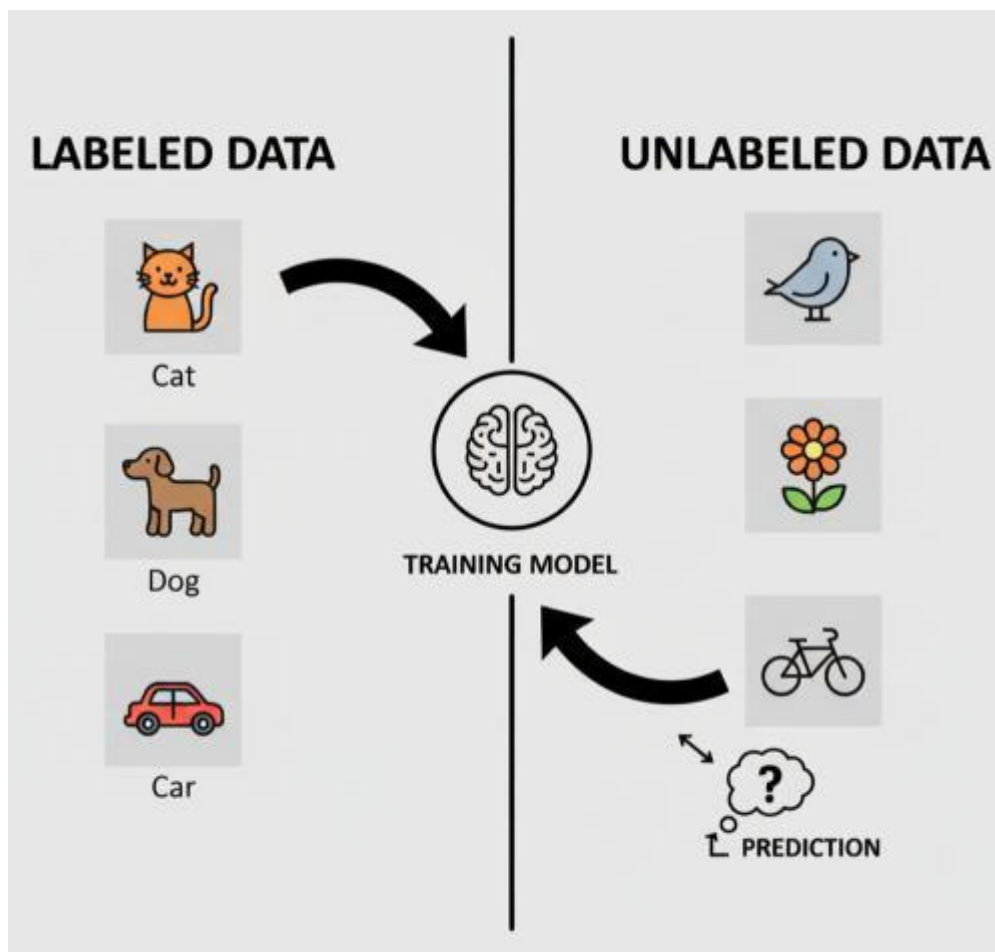
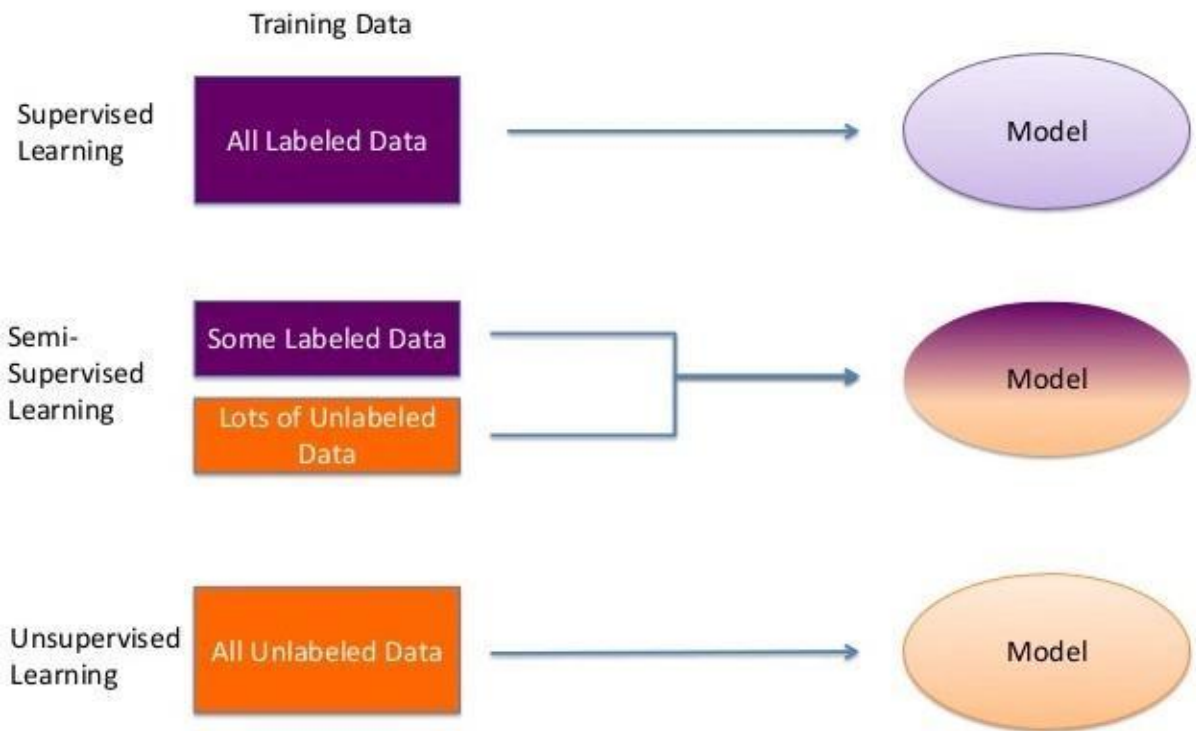


Tagged and used in supervised learning.

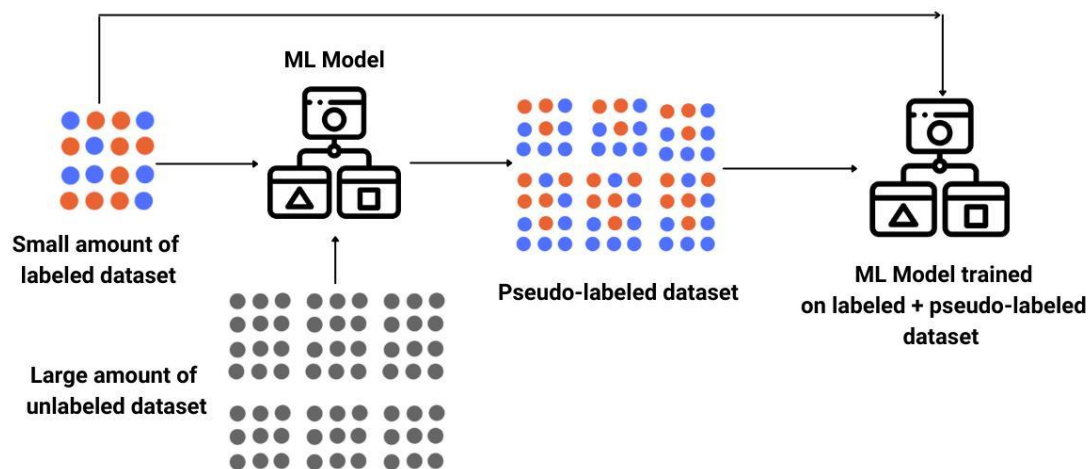
Unlabeled data



Untagged and used in unsupervised learning.



Semi-supervised learning use-case



Comparison of Types

Type	Data Used	Main Goal
Supervised Learning	Labeled Data	Predict output
Unsupervised Learning	Unlabeled Data	Find patterns
Reinforcement Learning	Rewards & Penalties	Learn best action
Semi-Supervised Learning	Mixed Data	Improve accuracy with less labeled data

Advantages of Machine Learning

- Automates tasks
- Improves accuracy
- Handles large data
- Learns and improves over time

Disadvantages

- Requires large data
- Can be expensive
- Sometimes difficult to understand predictions

- Needs powerful computers for complex tasks
-

Real-Life Examples

- Netflix recommendations
 - Google Translate
 - Face recognition
 - Chatbots
 - Online shopping suggestions
-

Conclusion

Machine Learning enables computers to learn from data and make intelligent decisions. Different types of machine learning are used depending on the nature of the problem and the available data.